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Materials that you may use during the exam:

1. One page (double sides) cheat sheet (A4). ONLY equations/formulas are allowed.
2. NOT open book
3. NOT open notes (including the printed handouts the instructors distributed in class)
4. NOT open to any other resources (including homework, quizzes, practical exercises).

** Violation of the above policy will result in significant grade deduction.*

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Short answer (3 pts each)

(a) What is the overall tray efficiency in a distillation tower?

$$\text{Overall Tray Efficiency} = \frac{\text{No. of Theoretical Trays}}{\text{No. of actual Trays}}$$

(b) What is the Morphree tray efficiency?

For vapor phase: $E_{MV} = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}}$ for liquid phase: $E_{ML} = \frac{x_n - x_{n-1}}{x_n^* - x_{n-1}}$

(c) What is the mass flux in SI units?

$$\text{kg s}^{-1} \text{ m}^{-2}$$

(d) What are the Fick's first and second laws of diffusion?

① $J = -D \frac{dc}{dx}$ ② $\frac{dc}{dt} = D \frac{d^2c}{dx^2}$

(e) What is the main assumption for McCabe-Theile method of calculation for distillation?

assumption Constant molar Overflow

10/10 A. (10 pts) Calculate the vapor and liquid compositions in equilibrium at 95 °C for benzene-toluene using the vapor pressure from the Table at 101.32 kPa.

Vapor-Pressure and Equilibrium-Mole-Fraction Data for Benzene-Toluene System					
Vapor Pressure					
Temperature		Benzene		Toluene	
K	°C	kPa	mm Hg	kPa	mm Hg
288.2	15	116.9	877	46.0	345
303.2	30	135.5	1016	54.0	405
308.2	35	155.7	1168	63.3	475
373.2	100	179.2	1344	74.3	557
378.2	105	204.2	1532	86.0	645
383.8	110.6	240.0	1800	101.32	760

95 °C → Table → $P_A = 155.7 \text{ kPa}$
 $P_B = 63.3 \text{ kPa}$

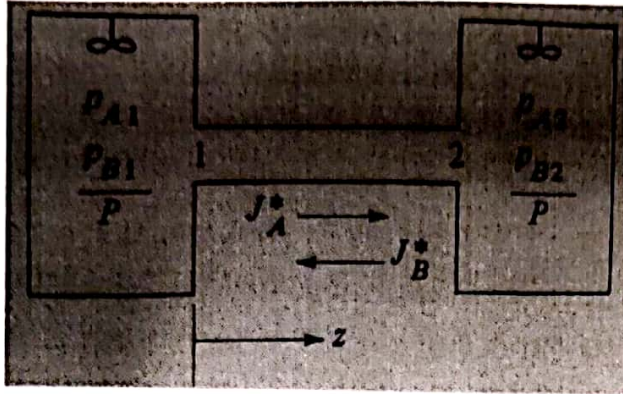
Raoult's law → $155.7(x_A) + 63.3(1-x_A) = 101.32 \text{ kPa}$
 → $x_A = 0.411$ → $x_B = 1 - x_A = 0.589$

$$y_A = \frac{P_A x_A}{P} = \frac{155.7(0.411)}{101.32} = 0.632$$

B. (10 pts) Equimolar counter diffusion: Ammonia gas (A) is diffusing through a uniform tube 0.10 m long containing N_2 gas at 1.0132×10^4 Pa pressure and 298 K. At point 1, $p_{A1} = 1.013 \times 10^4$ Pa and at point 2, $p_{A2} = 0.507 \times 10^4$ Pa. The diffusivity $D_{AB} = 0.230 \times 10^{-4} \text{ m}^2/\text{s}$.

(a) Calculate the flux J_A^* at steady state.

(b) Calculate the flux J_B^* at steady state.



$$J_A^* = \frac{D_{AB} (P_{A1} - P_{A2})}{RT(z_2 - z_1)} = \frac{(0.23 \times 10^{-4}) (1.013 \times 10^4 - 0.507 \times 10^4)}{8314 (298) (0.1 - 0)}$$

$$= 4.7 \times 10^{-7} \frac{\text{kg mol}}{\text{m}^2 \text{ s}}$$

$$J_B^* = \frac{D_{AB} (P_{B1} - P_{B2})}{RT(z_2 - z_1)} = \frac{(0.23 \times 10^{-4}) (9.119 \times 10^4 - 9.625 \times 10^4)}{8314 (298) (0.1 - 0)}$$

$$= -4.7 \times 10^{-7} \frac{\text{kg mol}}{\text{m}^2 \text{ s}}$$

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C. (15 pts) The air in a room is at 26.7 °C and a pressure of 101.325 kPa and contains water vapor with a partial pressure $p_A = 2.76$ kPa. Assume air has 21% O_2 with a molecular weight of 32 and 79 % N_2 with a molecular weight of 28. The vapor pressure of water at 101.325 kPa is 3.5 kPa. Calculate

- (a) Humidity, H
 (b) Saturation humidity, H_s , and percentage humidity, H_p ,
 (c) Percentage relative humidity, H_R

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$$H = \frac{18.02}{28.97} \left(\frac{P_A}{P - P_A} \right) = \frac{18.02(2.76)}{28.97(101.3 - 2.76)} = 0.01742 \frac{\text{kg H}_2\text{O}}{\text{kg air}}$$

2

$$H_s = \frac{18.02}{28.97} \frac{P_{As}}{P - P_{As}} = \frac{18.02 \times 3.5}{28.97(101.3 - 3.5)} = 0.022 \frac{\text{kg H}_2\text{O}}{\text{kg air}}$$

3

$$H_p = 100 \frac{H}{H_s} = \frac{100(0.01742)}{0.022} = 78.3 \%$$

$$\rightarrow H_R = 100 \frac{P_A}{P_{As}} = \frac{100(2.76)}{3.5} = 78.9\%$$

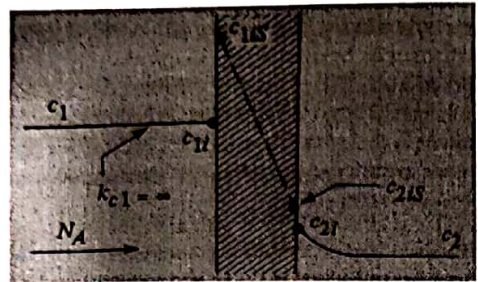
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D. (20 pts) A liquid containing dilute solute A at a concentration $c_1 = 3 \times 10^{-2} \text{ kmole/m}^3$ is flowing rapidly past a membrane of thickness $L = 3.0 \times 10^{-4} \text{ m}$. The distribution coefficient $K' = 1.5$ and $D_{AB} = 7.0 \times 10^{-11} \text{ m}^2/\text{s}$ in the membrane. The solute diffuses through the membrane, and its concentration on the other side is $c_2 = 0.5 \times 10^{-2} \text{ kmole/m}^3$. The mass-transfer coefficient k_{c1} is large and can be considered as infinite, and $k_{c2} = 2.0 \times 10^{-5} \text{ m/s}$.

(a) Derive the equation to calculate the steady state flux N_A .

(b) Calculate the flux and the concentrations (c_{1is} , c_{2is} and c_{2i}) at the membrane interfaces.



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$$C = C_{ii}$$

$$\text{Steady state flux} = N_A = \frac{C_1 - C_2}{\frac{1}{k_{c1}} + \frac{L}{D_{AB}K'} + \frac{1}{k_{c2}}} \xrightarrow{k_{c1} = \infty} = \frac{C_1 - C_2}{\frac{1}{D_{AB}K'} + \frac{1}{k_{c2}}}$$

2

$$\frac{1}{D_{AB}K'} = \frac{7 \times 10^{-11} (1.5)}{3 \times 10^{-5}} = 3.5 \times 10^{-6} \text{ m/s}$$

$$N_A = \frac{C_1 - C_2}{\frac{1}{D_{AB}K'} + \frac{1}{k_{c2}}} = \frac{3 \times 10^{-2} - 0.5 \times 10^{-2}}{3.5 \times 10^{-6} + \frac{1}{2.0 \times 10^{-5}}} = 7.458 \times 10^{-8} \text{ kgmol}/(\text{s} \cdot \text{m}^2)$$

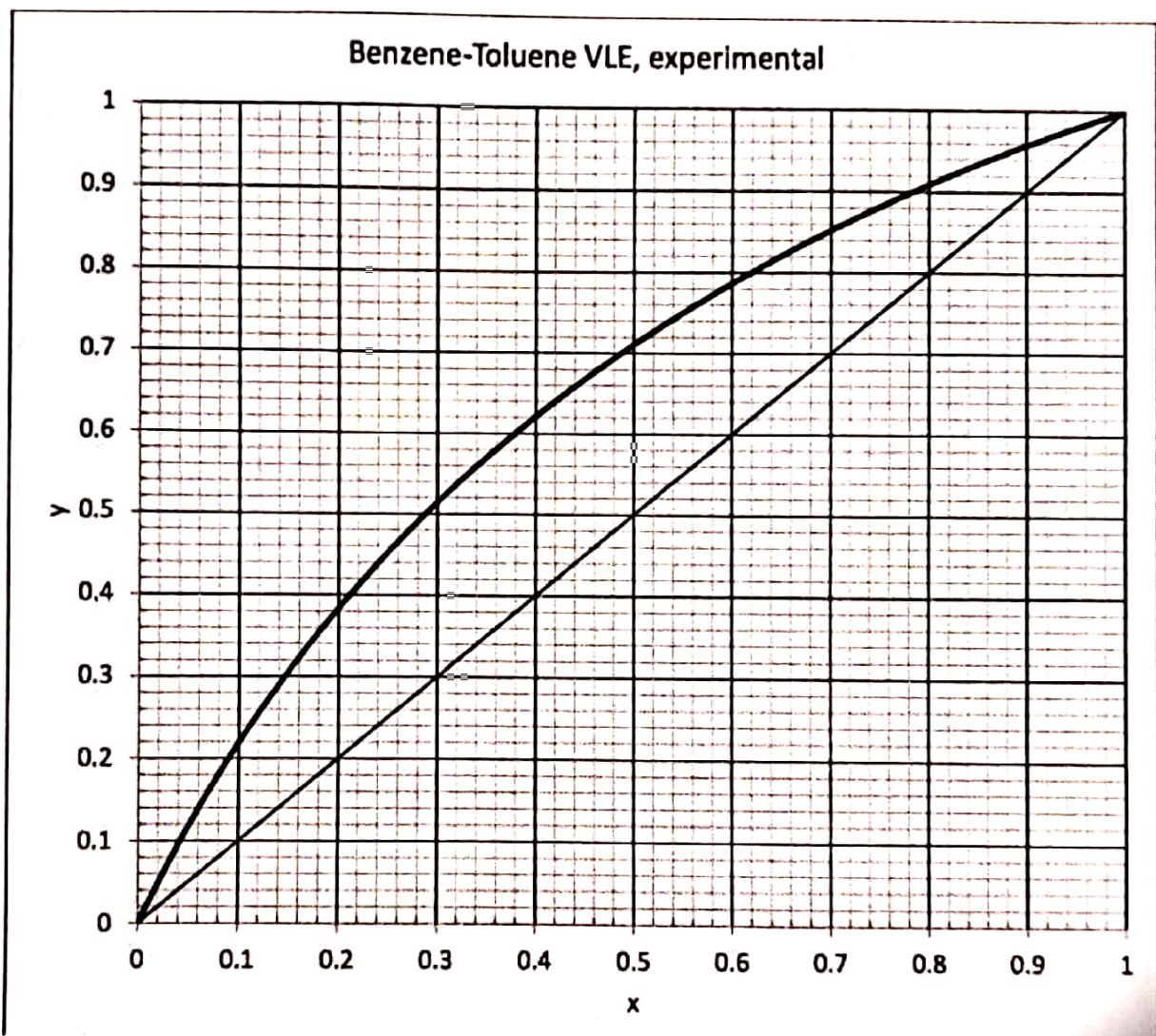
$$N_A = 7.458 \times 10^{-8} = k_{c2} (C_{2i} - C_2) = 2.0 \times 10^{-5} (C_{2i} - 0.5 \times 10^{-2})$$

$$C_{2i} = 0.869 \times 10^{-2} \text{ kgmol}/(\text{m}^3)$$

$$K' = 1.5 = \frac{C_{2is}}{C_{2i}} \rightarrow C_{2is} = 1.304 \times 10^{-2} \text{ kgmol}/(\text{m}^3)$$

E. (30 pts) A liquid mixture of benzene-toluene is to be distilled in a fractionating tower at 101.3 kPa pressure. The feed of 100 kmole/hr is saturated vapor, containing 45 mol% benzene and 55 mol% toluene, and enters at 366.7K. A distillate containing 95 mol% benzene and 5 mol% toluene and a bottom containing 10 mol% benzene and 90 mol% toluene are to be obtained. The reflux ratio is 4:1. The average heat capacity of the feed is 159 kJ/kmol.K and the average latent heat 32099 kJ/kmol. K. Calculate

- the kmole per hour distillate
- kmole per hour bottoms
- q value
- the number of theoretical trays needed
- the minimal number of trays for the total reflux



$$F = D + W \rightarrow Fx_f = Dx_D + Wx_W$$

$$100 = D + W \quad 100(0.45) = D(0.95) + (100-D)(0.1)$$

$$D = 41.2 \frac{\text{kg mol}}{\text{h}} \rightarrow W = 58.8 \frac{\text{kg mol}}{\text{h}}$$

$$y_{n+1} = \frac{R}{R+1} x_n + \frac{x_D}{R+1} = \frac{4}{4+1} x_n + \frac{0.95}{4+1} = 0.8 x_n + 0.19 \quad \text{enriching line}$$

$$H_v - H_L = 32099 \frac{\text{kJ}}{\text{kg mol}}$$

$$C_{pL} = 159 \frac{\text{kJ}}{\text{kg mol}}$$

$$\text{Boiling point of feed} = T_B = 366.7 \text{ K}$$

$$q = \frac{32099 + 159(366.7 - 327.6)}{32099} = 1.195$$

$$q = \frac{13800 + 38(200.3 - 130)}{13800} = 1.125 \quad \text{in English}$$

$$\text{Slope of } q \text{ line} = \frac{q}{q-1} = 6.12$$

5 Trays